Smaller Diophantine descriptions of ZFC consistency

Thirteen positive unknowns, a smaller quartic,
and a four-site computation compiler

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Abstract

We combine and extend the eight fixed-input constructions currently collected in ProveIt's *Con(ZFC) in Diophantine form* report. The main new equation has thirteen strictly positive integer unknowns, total degree 571, and an explicit 84-operation arithmetic circuit. Its unsolvability is equivalent, provably in ZFC, to Con(ZFC). A four-equation presentation uses 78 operations. The key algebraic step replaces three square-root witnesses by a single weighted sum, with a proof that the eliminated roots remain integers and that the resultant has no degenerate denominator branch.

Other combinations improve different measures: a fifteen-variable, six-equation system has degree 28 and 61 operations; its single product has degree 99 and 69 operations. A direct lift gives 23 quadratic equations in 32 positive unknowns and hence one quartic. An explicit domain refinement gives a quartic in 62 unrestricted integer unknowns. Finally, replacing nine-site windows by four-site windows reduces the local alphabet to 71 356 645 and the bit length of the distinguished coefficient $B-1$ from 5^{67} to 5^{28}, with the same four coefficient ports.

Every coefficient is fixed by a terminating finite recipe and the literal K215 table. The enormous coefficients are not expanded. The new results are ordinary mathematical proofs, accompanied by explicit circuits and finite algebra and compiler checks; they are not newly checked Lean or Rocq theorems. The constructions improve the reviewed collection in specified measures, without claiming a global minimum, a new universal operation bound, or a proof of ZFC consistency.

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1 Results and comparison

Write $\mathrm{Ref}_{\mathrm{ZFC}}$ for the arithmetic assertion that there exists a finite derivation of a contradiction from ZFC. Throughout, Con(ZFC) means $\neg\mathrm{Ref}_{\mathrm{ZFC}}$ for an ordinary effective presentation. The machine calibration and its internalization are discussed in Section 10.

Theorem 1.1 (Main result)**.** *Let (β, K_0, M_C, M_F) be the four fixed integers of Section 8, obtained from the literal blank-input K215 machine. For the polynomial P_{13} defined in Section 3,*

$$\mathrm{ZFC} \vdash \left(\exists \mathbf{x} \in \mathbb{Z}_{>0}^{13}\ P_{13}(\mathbf{x}) = 0\right) \iff \mathrm{Ref}_{\mathrm{ZFC}}.$$

Consequently,

$$\boxed{\mathrm{ZFC} \vdash \left(\forall \mathbf{x} \in \mathbb{Z}_{>0}^{13}\ P_{13}(\mathbf{x}) \neq 0\right) \iff \mathrm{Con(ZFC)}.}$$

The same assertion holds for each alternative explicitly defined below, in its stated domain. In particular there is a specified quartic in 32 positive unknowns and a specified quartic in 62 unrestricted integer unknowns with the corresponding equivalence.

The eight packages are numbered 01–08 in the repository's reconciliation note [1]. The best existing arithmetic baseline is the independently obtained 15-variable, 67-operation polynomial of 06 and 08. Its degree is 155. The four coefficients differ between those packages; the same circuit shape does not make their numerals interchangeable. We use 06's raw-K215 semantics as the base and prove a new coefficient recipe for it.

Construction	Positive variables	Equations	Degree	Operations
Existing 06/08 product	15	1	155	67
Existing 06 basic system	15	6	60	61
Existing 02 ordinary system	16	7	28	63
Existing 08 quadratic lift	34	25	2	—
New ordinary system	15	6	28	61
New ordinary product P_{15}	15	1	99	69
New retained-auxiliary system	16	7	24	60
New amalgamated system	14	5	72	67
New amalgamated product P_{14}	14	1	121	76
New weighted system	13	4	544	78
New weighted product P_{13}	13	1	571	84
New alternative-weight product	13	1	515	85
New quadratic lift	32	23	2	—
New quartic Q_{32}	32	1	4	—

For a system, "degree" means the maximum residual degree. Every displayed degree is exact. Arithmetic counts charge every binary $+$, $-$, or $\times$, including multiplication by a numeral. Reuse of an already computed register and the final equality comparison are free. The sum-of-squares alternatives are 78 operations/degree 56 for the ordinary fifteen-variable system, 80/degree 48 for the retained sixteen-variable system, 81/degree 144 for the fourteen-variable system, and 89/degree 1088 for the thirteen-variable system.

There is a genuine tradeoff. The new thirteen-variable equation uses fewer unknowns than all eight explicit packages, but does not beat the 67-operation evaluator. The new ordinary system lowers degree from 60 to 28 at the same variable and operation counts. The quartic reduces the positive-variable count from 34 to 32. In the unrestricted integer domain our 62-variable quartic improves the 83-variable quartic reported by 04; comparisons across different domains would be misleading. Classical nine-variable existence results remain relevant and prevent any claim that thirteen is an absolute minimum; the repository's `Jones1982.nine_unknowns` is an existence theorem, not an emitted polynomial with a stated degree [10].

2 Common polynomials and the inherited interface

All free variables in Sections 2–5 are strictly positive integers. A capitalized abbreviation does not introduce an additional existential variable. Define

$$\begin{aligned} q &= \beta J + 1, & C_0 &= q - F - \alpha, & X &= wq, & Y &= sq^3, \\ E &= XY, & k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\ H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, & D_0 &= X + ac + \gamma H. \end{aligned} \tag{1}$$

Put

$$\begin{aligned} G &= (q-1)(q-F) + \alpha, & R &= G(q^2-1) + (M_C + qM_F)J, \\ L_0 &= E(kY), & A &= 1 + L_0(L_0 + k), \\ N_1 &= D_0^2 - \Delta c^2, & N_k &= k - hE - R, \end{aligned} \tag{2}$$

$$N_t = (K_0 + w)C_0 + (q - F) - z(q-1). \tag{3}$$

Here z is the transport quotient, while T below is a different auxiliary quotient. The coefficient M_F is the *shifted* source mask, not the native one-cell high mask.

For comparison, package 06's normalized system uses the fifteen unknowns

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma$$

and, with

$$Q = (\Delta i c^2)^2, \qquad V = c(Tf - 1) - Rf^2,$$

has the six equations

$$\tau^2 = A, \quad N_1 = N_k = N_t = 1, \quad f^2 = 1 + \Delta i^2 c^4, \quad QV^2 - (Q-1)y^2 = 1. \tag{4}$$

This system is denoted $\mathcal{S}_{\mathrm{norm}}$. Package 06 proves that its positive solvability is equivalent to the blank halting of K215 [2]. Sections 8–9 prove that the new four-site coefficients satisfy the same compiler interface, so the same arithmetic theorem applies to them.

The following bounds will be used before any computation is decoded. Suppose $N_t \in \{1, -1\}$. Since

$$N_t = (K_0 + w)C_0 + 1 - F - (z-1)(q-1),$$

one has $C_0 \geq 0$; $C_0 \leq -1$ would make $N_t \leq -2$. One cannot yet assert $C_0 > 0$: the boundary $C_0 = 0, F = 2, z = 1$ permits $N_t = -1$. With the authentic mask bounds,

$$\begin{gathered} q \geq B \geq 16, \qquad q \leq G \leq q^2 - q, \\ 0 < (M_C + qM_F)J < (q-1)(1+2q), \\ q(q^2-1) < R < q^4 - q^3 + q^2 - 1 < q^4 - 2, \\ R > 3q + 1, \qquad E \geq q^4 > R + 2, \qquad a > R + 2. \end{gathered} \tag{5}$$

These follow directly from $q - F = C_0 + \alpha > 0$ and $0 < M_C < B - 1$, $0 < M_F < 2(B-1)$, $J = (q-1)/(B-1)$. If $N_k = \varepsilon \in \{1, -1\}$, then $k = R + \varepsilon + hE > R$. The literal split of c always gives

$$kY < c < k(Y+1). \tag{6}$$

3 The thirteen-variable equation

Supply exactly

$$\boxed{J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, \gamma, t \in \mathbb{Z}_{>0}.}$$

The three coordinates f, τ, y of (4) are absent. Using (1)–(3), define

$$\begin{aligned}
&Q = (\Delta i c^2)^2, && D = 1 + \Delta i^2 c^4, && v = cT, && z_0 = c + RD,\\
&d_0 = Q - 1, && M = D + v^2, && P = QDv^2, && Z = Qz_0^2,\\
&e = P + Z - 1, && L = d_0 t^2, && N = d_0 M^2 A, && C = L + N - e,\\
&a_0 = PZ, && b_0 = LN, && U = C^2 + 4(a_0 - b_0), &&\\
&\mathcal{R} = U^2 - 16C^2 a_0. &&&&&&
\end{aligned} \tag{7}$$

The requested single equation can be taken to be

$$\boxed{P_{13} = N_1 N_k N_t (1 + \mathcal{R}) - 1 = 0.} \tag{8}$$

An equivalent existence condition is the four-equation system

$$\boxed{N_1 = N_k = N_t = 1, \qquad U^2 = 16C^2 a_0.} \tag{9}$$

All symbols in these displays are ordinary polynomial abbreviations. There is no variable exponentiation, division, bit operation, primality test, or implicit program parameter in the evaluated polynomial. The proof uses rational expressions only to recover eliminated integer witnesses.

3.1 The weighted root lemma

Lemma 3.1 (Three roots represented by one integer)**.** *Let A, D, Q, v, z_0, M be positive integers with $A, D, Q > 1$, $v, z_0 > 0$, and put $d_0 = Q - 1$, $e = Q(Dv^2 + z_0^2) - 1$. Assume $d_0 M^2 A > e$. For $t > 0$, set*

$$C = d_0(t^2 + M^2 A) - e, \quad B_0 = 2Qvz_0, \quad U = C^2 + DB_0^2 - 4d_0^2 t^2 M^2 A.$$

Then

$$U^2 - D(2CB_0)^2 = 0$$

implies the existence of nonzero integers f, τ, y satisfying

$$f^2 = D, \quad \tau^2 = A, \quad Q(vf - z_0)^2 - (Q-1)y^2 = 1, \quad t = M\tau + y.$$

Conversely any such roots with $t = M\tau + y > 0$ give the resultant equation.

Proof. The hypothesis makes $C > 0$, so $2CB_0 \neq 0$. Define $f = -U/(2CB_0)$. The resultant gives $f^2 = D$. A rational number whose square is an integer is itself an integer: write it in lowest terms and compare the denominator in the squared equation. Thus f is an integer and is nonzero. Expanding $U + 2CB_0 f = 0$ gives

$$(C + B_0 f)^2 = 4d_0^2 t^2 M^2 A.$$

Hence $\tau = (C + B_0 f)/(2d_0 tM)$ is a rational number with square A, so it is a nonzero integer. Set $y = t - M\tau$. Then

$$d_0 y^2 = e - B_0 f = Q(vf - z_0)^2 - 1.$$

The value $y = 0$ would imply that a multiple of $Q > 1$ equals 1. The reverse implication follows by the same identities. No algebraic number field or unproved rational-to-integer step is needed. □

3.2 Why the chosen weight works

At a zero of (8), each of its four factors is an integer unit. Since Δ is 0 or 3 modulo 4, N_1 cannot equal -1. Also $\mathcal{R} \equiv U^2 \pmod 4$, so $1 + \mathcal{R}$ cannot equal -1. Therefore $N_1 = 1$, $\mathcal{R} = 0$, and $N_k = N_t = \varepsilon \in \{1, -1\}$. The weak bounds above are available at precisely this stage.

Write $U_0 = XY^2$. Then

$$A = 1 + k^2 U_0(U_0 + 1) > k^2 Y^4 > 8k^2((Y + 1)^2 + 1) > 8(c^2 + R^2).$$

The middle inequality holds for $Y \geq 4$, while here $Y \geq 4096$. For $M = D + v^2$,

$$M^2 \geq 4Dv^2, \qquad M^2 \geq D^2, \qquad z_0^2 = (c + RD)^2 \leq 2D^2(c^2 + R^2).$$

Splitting AM^2 into two halves proves $AM^2 > 2(Dv^2 + z_0^2)$. Since $Q \geq 2$ and $d_0 \geq Q/2$,

$$d_0 AM^2 > Q(Dv^2 + z_0^2) = e + 1. \tag{10}$$

Thus the lemma applies without already knowing the eliminated roots. The weight is substantive: an unweighted elimination can have a vanishing coefficient of f, so its resultant alone need not force D to be a square.

The shorter evaluator (7) is the lemma's resultant because $DB_0^2 = 4PZ$ and $d_0^2 t^2 M^2 A = LN$.

3.3 Soundness of the signed reconstructed roots

The lemma restores (4), with N_k, N_t still units of equal sign and with f, τ, y provisionally signed. Replace τ, y by their nonzero absolute values. If $f < 0$, replace both f, T by their negatives; every relevant expression uses f only through f^2 and Tf. This makes $f > 0$ while allowing T to be an arbitrary signed integer. It is essential to justify this domain extension instead of applying a positive-T theorem silently.

For an integer $A_0 \geq 2$, define the Pell sequences by

$$\chi_{A_0}(r) + \psi_{A_0}(r)\sqrt{A_0^2 - 1} = (A_0 + \sqrt{A_0^2 - 1})^r.$$

Use $A_0 = a + 2$ and $P_0 = 2XY^2 + 1$. The first and main norms imply

$$\tau = \chi_{P_0}(n), \quad k = 2\psi_{P_0}(n), \quad D_0 = \chi_{A_0}(p), \quad c = \psi_{A_0}(p).$$

The index congruence gives $2n \equiv R + \varepsilon \pmod E$. Since $0 < R + \varepsilon < E$, it follows that $n \geq (R - 1)/2 \geq 24$. Since $P_0 > A_0$ and $c > k$, Pell growth gives $p > n$ and $c > 2p$. Also $k > R$ implies $c > 2R$.

The normalized strong equation gives $f = \chi_{A_0}(m)$ and $\psi_{A_0}(m) = ic^2$. Strong divisibility of ψ implies $p \mid m$. Writing $m = pj$ and expanding $(D_0 + c\sqrt{\Delta})^j$ modulo c^2 gives

$$\psi_{A_0}(pj)/c \equiv jD_0^{j-1} \pmod c.$$

Because $\gcd(D_0, c) = 1$, one obtains $c \mid j$; hence $pc \mid m$ and $m > 2p$.

Put $S = \Delta i c^2$ and retain the sign of V. The auxiliary norm classifies $|SV| = \chi_S(\ell)$, $y = \psi_S(\ell)$ with odd $\ell = 2r + 1$. If $\chi_X(2r + 1) = XC_r(X^2)$, the recurrence gives

$$C_r(0) = (-1)^r \ell, \qquad C_r(-\Delta) = (-1)^r \psi_{A_0}(\ell).$$

The literal congruences $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$, together with $S^2 \equiv -\Delta \pmod f$, give $R \equiv \pm p \pmod c$ by the signed Pell step-down argument. For completeness, squaring the first congruence yields

$$\chi_{A_0}(2\ell) \equiv \chi_{A_0}(2p) \pmod{\chi_{A_0}(m)}.$$

Reduction of the index modulo $2m$, reflection, and $2\chi_{A_0}(m-1) < \chi_{A_0}(m)$ force $\ell \equiv \pm p \pmod{m}$. The second congruence then gives the claim modulo c. Since $0 < R, p < c/2$, this forces $p = R$. This argument uses $|V|$, and never assumes $T, V > 0$ [5, 6].

Now write $2n = R + \varepsilon + jE$ with $j \geq 0$. The bound $E > R + 2$ excludes $j \geq 1$, because $n < p = R$. If $\varepsilon = -1$, then $p = 2n + 1$. Pell duplication, with $\chi_{A_0}(2) > P_0$, would give $c = \psi_{A_0}(2n + 1) > A_0 k > k(Y + 1)$, contrary to (6). Thus $\varepsilon = 1$, and $C_0 > 0$ follows.

The remaining exterior argument uses no sign of V or T. The lower Pell-ratio estimate gives $2^R < a$; the main-root congruence modulo $H = 4a + 3$ gives $X \equiv 2^R \pmod{H}$. Since $0 < X, 2^R < a < H$, one has $X = 2^R$. Hence $q \mid X$ makes q a power of two; the repunit relation makes $q = B^n$ for some $n > 0$. The bound $R > 3q$ restores $q^3 \mid X$. The half-binomial extraction gives

$$\nu_2(Y) = \mathrm{pc}(R) - 2, \qquad \mathrm{pc}(R) \geq 3dn + 2.$$

The shifted-mask population identity supplies the reverse bound and therefore the exact low and high masks. Their native convolution, anchor alignment, and overlap constraints decode a blank halting tableau. These exterior arguments are stated with their sign-independent hypotheses in the inherited notes [6, 7, 2]; the new tableau and coefficient interface is proved below. No numerical-input Pell bridge is needed.

Conversely, a blank halting run supplies a positive solution of (4). Set $t = M\tau + y > 0$ in the lemma; then (9) and (8) hold. This proves both directions.

3.4 Degree and operation certificates

Expression	Degree	Expression	Degree
c, a, Δ, R	$5, 6, 12, 4$	A, Q, D, v, z_0	$22, 46, 34, 6, 38$
P, Z, M	$92, 122, 34$	L, N, C	$48, 136, 136$
a_0, b_0, U	$214, 184, 272$	$\mathcal{R}$	544
N_1, N_k, N_t	$18, 7, 2$	P_{13}	571

The highest homogeneous part of C comes uniquely from N; that of U comes from C^2; and that of $\mathcal{R}$ comes from U^2. All are nonzero for every $\beta > 0$. For N_1, canceling the apparent top degree in $D_0^2 - \Delta c^2$ leaves the nonzero degree-18 leading term $8\gamma a_{\mathrm{top}}^2 c_{\mathrm{top}}$. The product of nonzero leading forms in an integer polynomial ring is nonzero. Hence $\deg P_{13} = 18 + 7 + 2 + 544 = 571$ uniformly on the actual coefficient slice.

The archive emits all $84 = 49M + 35A$ live gates. Removing its final unit-product gates and comparing U^2 with $16C^2 a_0$ gives $78 = 46M + 32A$ operations for (9). Adding four squared residuals costs 11 more operations and gives degree 1088. Modular univariate diagnostics supplement these symbolic leading-form proofs; they are not the justification of uniform exact degree.

3.5 One extra operation lowers the degree to 515

There is a useful second weight. Put $I = ic^2$ and replace M in (7) by

$$M = kI^2 + T^2.$$

All other polynomial definitions remain the same. For efficient evaluation, compute $D = 1 + \Delta I^2$, $Q = \Delta(D - 1)$, and $v^2 = c^2 T^2$. These are identities with the earlier definitions, not new equations. The resulting product uses $85 = 50M + 35A$ operations and has degree 515; its four-equation system uses 79 operations with maximum degree 488. The sum of squared residuals uses 90 operations and has degree 976.

Here is the additional domination check. Write $A_0 = a + 2 < 2E$. The weak bounds give $k > E$, $c < 2kY$, $k > 48$, and

$$L_0 = kEY > A_0 c/4, \qquad L_0 > E^2 Y > 4A_0^2.$$

Thus $A > A_0^2 c^2/16$ and $A > 16A_0^4$. Since $I > 1$,

$$D < A_0^2 I^2, \qquad M^2 \geq 4kI^2 T^2, \qquad M \geq kI^2, \qquad z_0 < 2kA_0^2 I^2.$$

The first set of inequalities gives $AM^2 > 4Dv^2$; the second gives $AM^2 > 4z_0^2$. Hence the same strict bound $AM^2 > 2(Dv^2 + z_0^2)$ proves $C > 0$, so the whole restoration and signed soundness argument applies unchanged. Now $\deg M = 23$, $\deg N = 114$, while e has degree 122. The leading form of C is $-e_{\mathrm{top}}$; that of the resultant is C_{top}^4. Consequently its degree is 488 and the product's degree is $488 + 18 + 7 + 2 = 515$. The archive includes both weights; the first has the shorter proof and cheaper evaluator.

4 Lower degree with the ordinary strong equation

Package 02's ordinary strong equation can be combined with 06's inputless compiler. This changes the scaling of one witness, so the equivalence is an equivalence of existence conditions, not a coordinatewise identity. Retain the fifteen variables of (4) and set

$$I = ic^2, \qquad K = \Delta(f^2 - 1), \qquad V = c(Tf - 1) - Rf^2.$$

The new system is

$$\boxed{\tau^2 = A, \quad N_1 = N_k = N_t = 1, \quad I^2 = K, \quad K(V^2 - y^2) + y^2 = 1.} \tag{11}$$

It has six equations, with exact residual degrees $22, 18, 28, 7, 2, 22$ in first/main/auxiliary/index/-transport/strong order. Its shared evaluator has $61 = 34M + 27A$ operations. Define

$$N_0 = \tau^2 - L_0(L_0 + k), \quad N_a = K(V^2 - y^2) + y^2, \quad N_s = 1 + I^2 - K.$$

An equivalent single equation is

$$\boxed{P_{15} = N_0 N_1 N_a N_k N_t N_s - 1 = 0.} \tag{12}$$

It has $69 = 39M + 30A$ operations and exact degree 99. The sum of the six squared residuals has 78 operations and degree 56.

4.1 Why the relaxed equation is sufficient

Completeness is immediate from the normalized system: replace its i by Δi. Then the new $I^2 = \Delta(f^2 - 1)$ equals the old auxiliary coefficient $(\Delta ic^2)^2$, preserving all other coordinates and equations.

Soundness uses the following independently established interface [3, 4].

Lemma 4.1 (Ordinary strong rank)**.** *Let $A_0 \geq 2$, $\Delta = A_0^2 - 1$, $c = \psi_{A_0}(p)$, and $c > A_0\Delta^2$. If positive integers i, f satisfy $(ic^2)^2 = \Delta(f^2 - 1)$, there is $m > 0$ such that*

$$f = \chi_{A_0}(m), \quad ic^2 = \Delta\psi_{A_0}(m), \qquad p \mid m, \quad c \mid m.$$

In particular $c^2 \mid f^2 - 1$.

Proof mechanism. Factor Δ into its squarefree part times a square and use the fundamental norm-one unit in the integer-coefficient quadratic ring. If $A_0 = \chi_F(e)$ and $S = \psi_F(e)$, then $\Delta = (F^2-1)S^2$ and $\psi_F(ep) = Sc$. For the auxiliary index n, strong divisibility gives $h = \psi_F(\gcd(n, ep)) \geq c/\Delta$. If $\gcd(n, ep) < ep$, the proper-divisor bound and Pell doubling give $h^2 < A_0 c/2$, contradicting $c > A_0\Delta^2$. Thus $ep \mid n$, recovering an integral Pell solution at A_0. Writing $m = pj$, the binomial expansion modulo c yields $c \mid \Delta j$. Finally $\gcd(c, \Delta) = \gcd(p, \Delta) \mid p$, so $c \mid pj = m$. The full elementary proof, including squarefree divisibility and the fundamental-unit argument, is included in the archive [4]. □

The first/main norms and the index congruence establish $p > n \geq 24$, $c > (a+2)\Delta^2$, $c > 2p$, and $c > 2R$ before this lemma is used. This ordering matters: the individual relaxed norm can have extra rational Pell solutions without these hypotheses. The lemma gives $m \geq c > 2p$. It also gives $f > 2c$ and

$$K - (Rf^2 + c) = (\Delta - R)f^2 - \Delta - c > 0.$$

The auxiliary norm excludes $V = 0$. If $|V| > 1$, it implies $|V| \geq 2K - 1$ by comparing consecutive squares. But $T > 0$ gives $V > -Rf^2 - c > -K$, excluding negative V of magnitude greater than one. The congruence $V \equiv -c \pmod{f}$, with $f > 2c$, excludes $V = \pm 1$. Hence $V > 1$.

The integers

$$o = cT - Rf, \qquad j = Tf - 1 - R(f^2-1)/c$$

are positive because $of = V + c$ and $jc = V + R$. The quotient in the definition of j is integral by Lemma 4.1. The two congruences $V = of - c = jc - R$ give $p = R$ by the same auxiliary step-down, now with coefficient I^2. The exterior power, population, and tableau decoding follows as before.

For (12), first infer integer units. The consecutive-product negative Pell descent excludes $N_0 = -1$; modulo 4 excludes $N_1 = -1$. Since $K = \Delta(f^2-1)$ is 0 or 1 modulo 4, $N_s = 1 + I^2 - K$ cannot be -1. Thus $K = I^2$, and N_a is also a square modulo 4, so $N_a = 1$. The remaining equal signs on N_k, N_t are resolved with the weak $C_0 \geq 0$ bounds, ordinary rank, and the same index-sign argument used above. This proves product soundness without prematurely assuming $C_0 > 0$.

4.2 Retaining two auxiliary witnesses

Replace T by positive o, j, put $V = of - c$, and append $V = jc - R$ to (11). This gives sixteen variables and seven equations. The residual degrees are $22, 18, 24, 7, 2, 22, 6$; the evaluator costs $60 = 33M + 27A$ operations. Its soundness proof is shorter: once $f > 2c$, the inequality $V = of - c > c > 1$ is immediate. The sum of squares has degree 48 and 80 operations. These retained witnesses provide the useful starting point for the quartic.

5 A fourteen-variable intermediate option

The ordinary system also permits a less costly root combination. Remove τ, y and retain f, supplying one positive integer t. Let I, K, V, A have their ordinary meanings and put

$$d = K - 1, \quad e = KV^2 - 1, \quad L = dt^2, \quad N = dA, \quad u = L + N - e, \quad \mathcal{R}_{14} = u^2 - 4LN.$$

The five equations are

$$N_1 = N_k = N_t = 1, \qquad I^2 = K, \qquad u^2 = 4LN.$$

They use $67 = 38M + 29A$ operations and have maximum degree 72. At a solution $K = I^2 > 1$, and

$$\tau = \frac{u}{2dt}, \qquad \tau^2 = A, \qquad y = t - \tau$$

recover integral nonzero τ, y and $dy^2 = e$. Their absolute values give an ordinary-system solution. Conversely, $t = \tau + y$ gives the five equations.

The product

$$\boxed{P_{14} = N_1 N_k N_t (1 + I^2 - K)(1 + \mathcal{R}_{14}) - 1 = 0}$$

has $76 = 42M + 34A$ operations and degree 121. Both new unit factors exclude -1 modulo 4; therefore the same reconstruction and ordinary sign proof apply. Its sum-of-squares alternative has 81 operations and degree 144. The archive additionally records the inputless specialization of 05's f-elimination: fourteen-variable products at 73 operations/degree 342 and 75/degree 293. These offer other arithmetic tradeoffs, but are not the main result here.

6 An explicit quartic in 32 positive unknowns

Retain the sixteen primary unknowns

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma$$

and adjoin sixteen positive register unknowns

$$N, E, P, c, a, X, D_0, C_2, G, L, F_2, V, g, I, K, \Delta.$$

Only $q = \beta J + 1$, $k = \eta + \zeta$, $H = 4a + 3$, and $C_0 = q - F - \alpha$ remain abbreviations. They are affine in the unknowns. Here the symbol L means $q^2 - 1$, and $P = L_0 + k$; these register names are local to this section.

The following are the complete 23 quadratic equations:

$$\begin{aligned}
L &= q^2 - 1, & N &= q(L + 1),\\
X &= wq, & a &= E + sN,\\
E &= X(a - E), & c &= k(a - E) + \eta,\\
P &= E(c - \eta) + k, & \tau^2 &= P(P - k) + 1,\\
D_0 &= X + ac + \gamma H, & \Delta &= a^2 + H,\\
C_2 &= c^2, & D_0^2 &= \Delta C_2 + 1,\\
k &= jc - V + 1 + hE, & G &= (q - 1)(q - F) + \alpha,\\
jc - V &= LG + (M_C + qM_F)J,\\
(K_0 + w)C_0 + (q - F) - z(q - 1) &= 1,\\
F_2 &= f^2, & V &= of - c,\\
g &= y^2 - V^2, & I &= iC_2,\\
K &= I^2, & K &= \Delta(F_2 - 1),\\
y^2 - Kg &= 1.
\end{aligned} \tag{13}$$

Let $r_1, \ldots, r_{23}$ be their left sides minus right sides. Then

$$\boxed{Q_{32} = \sum_{j=1}^{23} r_j^2 = 0.} \tag{14}$$

This is a single polynomial of exact total degree 4. Only the first Pell residual contains τ, with coefficient 1 on τ^2, so the coefficient of τ^4 in Q_{32} is 1.

To project a solution, set $Y = a - E = sq^3 > 0$ and $R = jc - V$. The packing equation forces the original expression for R; the transport equation forces $C_0 > 0$, hence $R > 0$. Every equation

of the retained-auxiliary system is recovered. Conversely its positive solutions extend to all sixteen registers. For the less immediate sign, the auxiliary norm gives

$$(K-1)(y^2-V^2)=V^2-1>0,$$

so $g>0$. All other register signs follow from their displayed definitions and the earlier budget bounds. Thus positive solvability is equivalent in both directions.

The two-register saving over 08 has a specific explanation. Ordinary scaling removes the separate register for Δc^2 used by the normalized strong multiplier. Replacing T by o, j costs one primary variable but removes the register $Tf-1$. Finally, $R=jc-V$ removes the packed-index register and its defining equation without increasing any residual's degree. The final count is 32 unknowns and 23 quadratics instead of 34 and 25.

7 Domains: natural numbers and unrestricted integers

7.1 Natural-number forms

To obtain a polynomial over $\mathbb{N}=\{0,1,2,\dots\}$, replace every positive coordinate x by $x+1$. This preserves the number of unknowns and the exact degree. It changes the displayed evaluation cost if the extra additions are charged; no positive-domain count is silently reused for the shifted evaluator.

7.2 A quartic in 62 unrestricted integer unknowns

In (13), positivity need be explicitly imposed only on the following ten primary coordinates:

$$\mathcal{P}=\{J,F,\alpha,z,i,s,w,\eta,\zeta,\gamma\}. \tag{15}$$

The six coordinates f,h,o,j,τ,y and all sixteen registers can be allowed to range over $\mathbb{Z}$. This relaxation requires a soundness proof; it is not a rule that arbitrary auxiliary signs may be discarded.

With (15), the quadratic definitions force $q,X,Y,E,k,c,a,\Delta,D_0,I,K>0$ and $K>1$. The transport equation forces $C_0>0$, giving $0<R<E-2$. Since $k=R+1+hE>0$, an integer $h\le -1$ is impossible. Thus $h\ge 0$ and $k>R$, which are all that the rank bootstrap needs. The first norm makes $\tau\neq 0$, and ordinary strong scaling makes $|f|>1$. Apply Lemma 4.1 to $|f|$. The auxiliary norm makes $y,V\neq 0$ and gives the two congruences

$$V\equiv -c \pmod{|f|}, \qquad V\equiv -R \pmod{c}.$$

Classifying the absolute auxiliary Pell root gives $p=R$ with exactly the signed step-down used in Section 3. Neither o nor j needs a sign in these congruences. The no-wrap argument gives $2n=R+1$; in fact $k=2\psi_{P_0}(n)>2n=R+1$ then restores $h>0$.

For exterior decoding, use only the forward ratio, half-binomial extraction, and valuation statements that depend on $p=R$, $2n=R+1$, and positive X,Y,c,k. They do not require $V>0$ or a prior assertion $R\equiv 3 \pmod 4$. The resulting population threshold forces the native masks and hence the same blank tableau. One must not invoke the complete positive-domain kernel on this signed tuple without this qualification. Positive completeness is unchanged.

Every positive integer has the division-free representation

$$\mathrm{Pos}(u,v,w)=1+u^2+v^2+w(w+1), \qquad u,v,w\in\mathbb{Z}. \tag{16}$$

Its values are positive. For surjectivity, the three-square theorem applied to $4n-3\equiv 1 \pmod 4$ gives one odd square and two even squares; write them as $(2w+1)^2,(2u)^2,(2v)^2$. For each $x\in\mathcal{P}$, append the quadratic equation

$$x-\mathrm{Pos}(u_x,v_x,w_x)=0.$$

There are $32 + 3 \cdot 10 = 62$ unrestricted integer unknowns and $23 + 10 = 33$ quadratic equations. Define

$$\boxed{Q_{62,\mathbb{Z}} = \sum_{j=1}^{23} r_j^2 + \sum_{x \in \mathcal{P}} \big(x - 1 - u_x^2 - v_x^2 - w_x(w_x + 1)\big)^2 = 0.} \tag{17}$$

This is an ordinary integer polynomial of exact degree 4; the τ^4 coefficient is still 1. The preceding signed-domain argument proves

$$\mathrm{ZFC} \vdash \big(\neg\exists \mathbf{z} \in \mathbb{Z}^{62}\ Q_{62,\mathbb{Z}}(\mathbf{z}) = 0\big) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

7.3 A 39-variable integer version of the main equation

Substituting a separate copy of (16) for each of the thirteen positive coordinates of P_{13} gives a specified polynomial in 39 unrestricted integer variables. Its degree is exactly 1142, and its evaluator has $84 + 13 \cdot 7 = 175$ operations. Each substitution uses three multiplications and four additions. Nonvanishing of the highest homogeneous part follows by setting the second and third coordinates of each block to zero; the remaining substitution $x_j \mapsto u_j^2$ is injective on monomials. This offers fewer unrestricted unknowns than the 41-variable integer products of 06, at a larger degree.

8 Fixed coefficients from four-site windows

This section specifies the four integers used by every new construction. Only coefficient construction uses finite enumerations, exponentiation, and bit lengths. Once constructed, these are fixed integers, not extra unknowns or additional operations in the displayed polynomial circuit.

8.1 The exact machine and tile relation

Use the literal `compiler/k215.tm` supplied with this archive. Its SHA-256 digest is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

The source has 215 working states $0, \ldots, 214$, binary symbols, and halt state 215. Complete the halt state by staying put and preserving its scanned symbol. The first transition is $(0, 0) \mapsto (93, 1, \mathrm{right})$. This is the raw machine calibrated in the K215 endpoint, with no added input or initialization machine [9, 2].

Let $a_t = 439$. The tile alphabet has headless bits $0, 1$, head-bearing tiles $2(s + 1) + b$ for $0 \le s \le 215$ and $b \in \{0, 1\}$, and

$$L = 434, \quad Q_{\mathrm{init}} = 435, \quad R_{\mathrm{init}} = 436, \quad V_{\mathrm{bdry}} = 437, \quad H_{\mathrm{bdry}} = 438.$$

The three phase tiles have blank payloads, with an initial state-0 head only at Q_{init}. V_{bdry} has the vertical boundary flag; H_{bdry} has the horizontal boundary flag. All other tiles have neither flag.

The original nine-site local relation inspects only its cross (N, W, C, E, S). Moreover, its only use of N is the initialization presence test: an interior center has a phase if and only if N is a horizontal boundary. Translate that test one time step forward: an interior S has a phase if and only if C is a horizontal boundary. All other rules can be stated on the four-site tuple (W, C, E, S):

1. The vertical flag of C equals that of S. Two horizontally adjacent vertical boundaries and two temporally adjacent horizontal boundaries are forbidden. Away from vertical boundaries the horizontal flag propagates to the right.

2. If S is interior, its phase is present exactly when C is a horizontal boundary. After this test, a boundary center has no further center rule.

3. At a phase center, a left vertical boundary requires phase L, and a right vertical boundary requires phase R_{init}. If the right neighbor is not a vertical boundary, the phase pair is one of $(L, L), (L, Q_{\text{init}}), (Q_{\text{init}}, R_{\text{init}}), (R_{\text{init}}, R_{\text{init}})$.

4. A head immediately before a horizontal boundary must be halted. A moving head cannot escape through a neighboring vertical boundary.

5. When S is not a horizontal boundary, its payload is the deterministic successor of the three payloads at W, C, E. Vertical boundary neighbors count as headless blanks; other neighbors must have payloads. A center writes according to its transition; incoming heads from either neighbor and a staying center head are combined only if at most one head arrives.

The executable predicate is `compiler/tshape_helical.py:local_valid`. It implements exactly these clauses, using the literal completed table.

Lemma 8.1 (Translation of one local test)**.** *A tile plane satisfies the four-site relation at every center if and only if it satisfies the original nine-site relation at every center. The statement holds on the full integer plane and on every periodic plane, including presentations whose spatial and temporal shifts are not independent.*

Proof. The four corners of the original window are unused. All remaining clauses are unchanged except for the single presence test. Temporal translation is a bijection of either plane, so requiring that test at every center is equivalent to requiring its translate at every center. No primitivity or distinct-offset assumption enters. □

The distinguished Start tuple and the excluded tuple are

$$U_0 = (434, 435, 436, 1), \qquad U_1 = (437, 438, 437, 435).$$

Both satisfy their own local predicate. Start is forced at every initialized head, by its phase neighbors and the literal first transition. The excluded tuple has no global extension: its left vertical boundary propagates to the following row, where the following initialized Q_{init} would have a vertical boundary on its left, violating the phase endpoint rule.

8.2 Exact local alphabet count

There are 349 symbols sending a head from the left, including Q_{init}, and 82 sending one from the right. Among the 438 non-horizontal-boundary symbols this leaves 89 and 356 nonincoming choices, respectively. The following census is exhaustive.

Center type	Multiplicity	Valid tuples per center
Vertical boundary	1	192 282
Horizontal boundary	1	2 634
Headless bit	2	355 508
Halted head	2	223 966
Working head	430	162 788
Initial L	1	1 754
Initial Q_{init}	1	437
Initial R_{init}	1	1 750

For example the headless count is

$$439 \cdot 438 + 89 \cdot 356 + 349 \cdot 356 + 82 \cdot 89.$$

The first term ends the row at a horizontal boundary; the other terms allow zero or one incoming head and then determine the next payload. For a working head, its escape direction excludes one boundary choice. The left-moving count is

$$(89-1)356 + 349 \cdot 356 + 82(89-1) = 162\,788,$$

and the right-moving count has the same value. The phase pair graph and the endpoint rules give the last three rows. Thus the alphabet of valid four-site tuples has exactly

$$\boxed{\kappa = 71\,356\,645.}$$

The archive independently checks the formulas against the actual predicate for every left/right pair in each of thirteen exhaustive center categories. Enumerate valid tuples with U_0 first, U_1 second, and all remaining tuples in lexicographic order. This defines their selector numbers without any arbitrary choice.

8.3 Three overlaps suffice

For selected tuples U_i, indexed cyclically by $\mathbb{Z}/N\mathbb{Z}$, impose

$$U_i.E = U_{i-1}.C, \qquad U_i.C = U_{i-1}.W, \qquad U_i.S = U_{i-h}.C. \tag{18}$$

Putting $a_i = U_i.C$ recovers

$$U_i = (a_{i+1}, a_i, a_{i-1}, a_{i-h}).$$

The pullback plane at spatial/time coordinates (x, y) is a_{-x-hy}. Every actual neighborhood is therefore valid. This works for arbitrary N, h, with no assumption that $h \mid N$ or $\gcd(N, h) = 1$. Start rules out $h = 0$ modulo N because its center and following symbols differ.

Lemma 8.1 now gives the same finite-strip soundness as 06. Scanning horizontally from an initialized head finds vertical boundaries on both sides; the phase chains cannot pass through that head and close. The boundary tracks persist in time. In the finite cyclic time orbit, an initializing horizontal boundary eventually recurs; the first recurrence ends a genuine bounded blank computation, whose last head must be halted. Conversely a finite halting run admits the independently padded helical presentations of 06. Their Start and absence of the excluded selector survive the projection [2, 8].

8.4 Complete finite recipe for the four integers

Use the spatial payload set and copy-target sets

$$\mathcal{P} = ((1,0),(1,1),(1,2),(2,1)), \qquad \mathcal{V} = ((1,1)), \qquad \mathcal{H} = ((1,0),(1,1)).$$

Here $\mathcal{P}$ is ordered as written. For a positive integer x, $\mathrm{bits}(x)$ is its binary length. Put

$$\begin{aligned}
\ell &= \max(2, \mathrm{bits}(\kappa+1)), & A_b &= 2^\ell, \\
p_d &= \max\left(0, \left\lceil \frac{\kappa - 2\ell - 7a_t - 4}{2} \right\rceil\right), & j_{\max} &= 4a_t + 4 + p_d, \\
\mu &= A_b - 2 + \sum_{j=1}^{j_{\max}} A_b^j, & m_0 &= 2\ell + 11a_t + 8 + 2p_d, \\
n_{\mathrm{pos}} &= m_0 + 1, & n_{\mathrm{dum}} &= n_{\mathrm{pos}} - \kappa - 4a_t - 4, \\
e_* &= \kappa + 12a_t, & M_0 &= \kappa + 15a_t + n_{\mathrm{dum}}, \\
E_{\max} &= 27M_0, & H_c &= E_{\max} + 24M_0 + 3a_t + 1, \\
T_1 &= H_c + 2E_{\max} + a_t + 1, & T_2 &= T_1 + 4E_{\max} + 2.
\end{aligned}$$

Let L_{cell} be the least power of 5 at least $T_2 + 2E_{\max} + 2$. For $(r, c) \in \mathcal{P}$ and $0 \le u < a_t$, put

$$e(r,c,u) = \kappa + 3a_t + (2-r)3a_t + (2-c)a_t + u,$$
$$j(r,c,u) = 1 + a_t \operatorname{index}_{\mathcal{P}}(r,c) + u,$$

with indices starting at zero. Form the disjoint set $\mathcal{E}$ of native positions from the κ selectors $0, \dots, \kappa - 1$, the $4a_t$ retained payload positions, n_{dum} consecutive dummy positions beginning at e_*, and anchors $M_0, 3M_0, 9M_0, 27M_0$.

At each $e \in \mathcal{E}$ give a clause coefficient c_e as follows:

$$c_i = 1 + \sum_{(r,c)\in\mathcal{P}} A_b^{j(r,c,U_i[r,c])} + \sum_{t=1}^{4} A_b^{4a_t+t} \qquad (0 \le i < \kappa),$$
$$c_{e(r,c,u)} = A_b^{j(r,c,u)}, \qquad c_{3^{t-1}M_0} = A_b^{4a_t+t} \quad (1 \le t \le 4),$$
$$c_e = 0 \qquad \text{at a dummy position.}$$

Here $U_i[r,c]$ is its corresponding W, C, E, S component. Define

$$U_{\mathrm{bd}} = 4(n_{\mathrm{pos}} + 2)\big(28(\kappa + 4a_t + 4) + 6\big) + 8,$$
$$b_{\min} = \max\big(\ell(j_{\max} + 1) + 3,\ \ell(4a_t + 4) + \mathrm{bits}(U_{\mathrm{bd}}) + 1,\ 4\big).$$

Let b be the least power of 5 at least $b_{\min}$ and put

$$\rho = 2^b, \qquad d = bL_{\mathrm{cell}}, \qquad B = \rho^{L_{\mathrm{cell}}} = 2^d.$$

Finally set

$$D_C = \rho^{3a_t} + \rho^{H_c + a_t} + \rho^{8M_0} + \rho^{24M_0} + \sum_{e\in\mathcal{E}} c_e\big(\rho^{T_1 - e} + \rho^{T_2 - e}\big),$$
$$D_R = \rho^{H_c},$$
$$M_{F,0} = \mu(\rho^{T_1} + \rho^{T_2}) + \sum_{\substack{(r,c)\in\mathcal{V}\\ 0\le u<a_t}} \rho^{e(r,c,u)} + \sum_{\substack{(r,c)\in\mathcal{H}\\ 0\le u<a_t}} \rho^{H_c + e(r,c,u)} + \rho^{9M_0} + \rho^{27M_0},$$
$$M_C = B - 1 - \sum_{\substack{e\in\mathcal{E}\\ e\ne 1}} \rho^e - 2\rho^{e_*}, \qquad M_{F,\mathrm{native}} = M_{F,0} + 4. \tag{19}$$

The promised four ports are

$$\boxed{\beta = B - 1, \qquad K_0 = D_C + BD_R, \qquad M_C, \qquad M_F = M_{F,\mathrm{native}} + B - 1.} \tag{20}$$

This is a closed specification of four particular natural numbers. It uses the complete finite transition table and finite alphabet only; it does not run K215 or consult whether ZFC is consistent.

For clarity, the geometry evaluates to

Quantity	Exact value	Quantity	Exact value
ℓ	27	p_d	35 676 757
$j_{\max}$	35 678 517	m_0	71 358 405
n_{pos}	71 358 406	n_{dum}	1
M_0	71 363 231	$E_{\max}$	1 926 807 237
H_c	3 639 526 099	T_1	7 493 141 013
T_2	15 200 369 963	$b_{\min}$	963 319 989
b	5^{13}	L_{cell}	5^{15}

In particular,

$$\boxed{d = 5^{28} = 37\,252\,902\,984\,619\,140\,625.} \tag{21}$$

The exact inequalities

$$5^{12} < 963\,319\,989 \le 5^{13}, \qquad 5^{14} < 19\,053\,984\,439 \le 5^{15}$$

verify the two least-power choices. The integers are intentionally specified by recipe: expanding even $B-1$ would require d bits.

9 Why the smaller coefficient recipe is sound

Merely checking mask congruences is insufficient to replace the compiler constants. We verify the actual supports, clause fields, alignment, and completeness hypotheses of 06. The full parent compiler proof and helical proof are included under `upstream`; the following identifies every changed component.

9.1 Clause matching, copy fields, and mask balance

The payloads retain their former spatial exponents. At a vertical target $e(r,c,u)$, multiplication by ρ^{3a_t} selects exactly $e(r+1,c,u)$. At a horizontal target $H_c + e(r,c,u)$, multiplication by $\rho^{H_c+a_t}$ selects exactly $e(r,c+1,u)$, whereas D_R selects the next window's $e(r,c,u)$. All three source/target pairs belong to $\mathcal{P}$. Thus these are precisely the three equalities in (18). The selectors stop below κ; payloads stop below $\kappa + 12a_t$; dummies lie above the payloads. None creates an extra source at a retained copy target.

Compact clause indices remove only unused fields. A selected tuple supplies one selector, its four matching tile indicators, and the four anchors. The same two repeated clause bands therefore enforce exact selector/payload agreement. There are $3a_t$ copy parity tests, and

$$\mathrm{pc}(\mu) = \ell + 4a_t + 3 + p_d, \qquad \mathrm{pc}(M_{F,0}) = 2\,\mathrm{pc}(\mu) + 3a_t + 2 = m_0.$$

The native-position census $n_{\mathrm{pos}} = m_0 + 1$ gives exactly

$$\begin{gathered} \mathrm{pc}(M_C) + \mathrm{pc}(M_{F,\mathrm{native}}) = d, \qquad 0 < M_C, M_{F,\mathrm{native}} < B-1, \\ M_C \equiv 2 \pmod 4, \qquad M_{F,\mathrm{native}} \equiv 4 \pmod 8. \end{gathered} \tag{22}$$

These are equalities about the newly constructed masks, not hypotheses checked only on example masks.

9.2 Carry control and synchronization

The coefficient mass estimate used in the parent proof becomes

$$\sum_{e \in \mathcal{E}} c_e \le 14(\kappa + 4a_t + 4) A_b^{4a_t+4}.$$

There are at most κ selector terms, $4a_t$ payload terms, and four anchor terms; bounding each by its positive geometric sum gives the displayed estimate. The choice of $b_{\min}$ supplies the same quarter-radix carry margin as in 06.

Every native position is in $[0, E_{\max}]$. The inequalities

$$T_2 - T_1 > 2E_{\max}, \quad L_{\mathrm{cell}} - (T_2 - T_1) > E_{\max}, \quad T_2 + E_{\max} < L_{\mathrm{cell}}$$

separate the two clause bands and their rotations. Also $e_* + 1 < E_{\max}$. The clean-band argument consequently synchronizes a transport field to a whole-cell temporal shift before any copy equality is decoded. The anchor pair $9M_0, 27M_0$ still has the required unique difference $18M_0$. Omitting corner and north payloads changes neither assertion. A selected tuple has one tile indicator at every retained position; an empty cell has none. The horizontal E/C copy equality propagates occupancy from the marked Start throughout the cyclic word.

The sparse checker evaluates these support and coefficient statements for every native basis bit at every tested position in its finite fixtures. The general argument above supplies their parameter-uniform justification; small fixtures do not certify an enormous enumeration.

9.3 Actual-index completeness

The five-adic correction remains exactly

$$2D_C - D_R \equiv 3 \pmod 5.$$

Indeed $T_2 - T_1 = 4E_{\max} + 2$ makes the two clause-band sums cancel modulo 5. The four unpaired monomials, $a_t = 439 \equiv 3 \pmod 4$, and $\rho \equiv 2 \pmod 5$ give the displayed residue.

Hence the parent's ignored-dummy argument still adjusts the *actual final* packed index to the desired temporal stride. The one dummy position supplies an independently selectable bit in each word cell. Its Boolean subset argument uses the same unit coefficient modulo the five-power padding modulus. Canonical helical tableaux have one Start and nonzero temporal stride; these ensure the exceptional origin mask test. The dummies remain above the low selector fields. The same canonical carry estimate gives $2C_0 + F < q/3$, so $\alpha = q - C_0 - F > 0$. Only after this adjustment is the positive Pell converse invoked, at that final index. This avoids assuming a converse at an index that was later changed.

Thus both directions of the genuine native-compiler interface hold with (20). Composing it with the arithmetic proofs establishes the new constructions. The intermediate five-site cross compiler, also included in the archive, needs no translated test and gives $\kappa = 38\,256\,607\,897$ and $d = 5^{36}$. The original nine-site alphabet is exactly the cross alphabet times 439^4, because its four corners are unconstrained.

Window	Local alphabet size	Payload groups	bits$(B-1)$
Original nine-site	1 420 903 358 357 108 792 377	9	5^{67}
Five-site cross	38 256 607 897	5	5^{36}
New four-site	71 356 645	4	5^{28}

The final reduction in the distinguished coefficient's bit length is a factor of 5^{39}. It remains an enormous constant. This is a reduction in coefficient height, not a practical claim that the expanded polynomial can be stored.

10 The ZFC equivalence and its verification status

10.1 Logical calibration

At the pinned snapshot, the Lean source `KEndpoint215.lean` states the exact endpoint

```
theorem blank_halts_iff_zfc_refutation :
  BlankHalts machine215 <-> BProv ZFCax_s [] Form.fBot
```

and obtains its nonhalting counterpart by taking negations [9]. Here `BProv` names the repository's finitary proof relation; this endpoint is not a bound on the length of the refutation. The formal source calibrates the raw table used here.

All the newly used arguments are ordinary finite mathematics available in ZFC: arithmetic circuits, finite tableaux, Pell recurrences, elementary divisibility and bounds, finite Boolean subset constructions, and the classical three-square theorem for the integer-domain conversion. Their assertions about witness existence, integer signs, and the fixed machine can be arithmetized inside ZFC. The machine simulation and proof enumeration establish the ordinary effective equivalence between blank halting and existence of a finite ZFC refutation. Changing between the repository's axiom enumeration and a conventional effective presentation of ZFC does not change that consistency statement: the finitary proof translations are provable in ZFC.

The proof is therefore the following chain, in the stated domain:

$$\begin{aligned} \exists \mathbf{x}\ P(\mathbf{x}) = 0 &\longleftrightarrow \text{the four-site native masks encode a valid blank tableau} \\ &\longleftrightarrow \text{BlankHalts(K215)} \\ &\longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}. \end{aligned}$$

Taking negations yields Theorem 1.1. This does not infer ZFC provability from the soundness of a proof assistant: the mathematical reductions themselves are available in ZFC. The inherited formal endpoint documents the machine calibration; it is not a newly compiled formalization of the changed Diophantine reductions.

There is no claim here to prove Con(ZFC). Subject to the usual consistency assumption and the standard hypotheses of the second incompleteness theorem, ZFC cannot prove the asserted absence of solutions. The equivalence is a theorem about which arithmetic assertion expresses that consistency statement. No accepting run of the literal K215 machine, and no solution of the final equation, was found or used.

10.2 What was checked, and what remains mathematical

The archive makes the following distinctions explicit.

1. **Inherited formal endpoint.** The exact table and the statement of the Lean endpoint were inspected at the pinned commit. The whole Lean/Rocq project was not rebuilt in this work, and no claim of a fresh kernel check is made.
2. **New mathematical arguments.** Weighted root restoration, the signed quotient extension, the ordinary-strong combination, the smaller quadratic lift, the ten-coordinate positivity reduction, and both smaller neighborhood compilers are proved in this report and supporting notes. The delicate denominator and signed-domain arguments received independent review.
3. **Exact finite computation.** Standard-library Python scripts check arithmetic-circuit dependency graphs, full register liveness, operation counts, exact sparse identities and residual degrees, the literal transition-table hash, the local alphabet counts by exhaustive center categories, and sparse native layout identities.
4. **Supplementary diagnostics.** Random signed assignments and modular line restrictions check emitted circuits against independently written formulas. Toy halting tableaux, mutated periodic planes, and isolated root-restoration examples check implementations. They are not substitutes for the general proofs or witnesses for the final consistency equation.

The ordinary strong rank lemma and the native mask compiler are substantial inherited dependencies. Their precise hypotheses and uses are recorded here and in the included upstream audit notes. The new results should be reviewed as a mathematical contribution with reproducible checks, not advertised as already formalized.

11 Archive layout and reproduction

The root `README.md` gives the recommended integration entry points. The principal artifacts are:

- `con_zfc_reductions.tex` and its compiled PDF: this report.
- `arithmetic/polynomial13_84.json` and `arithmetic/polynomial13_lowweight85.json`: the complete live circuits for the two thirteen-variable equations.
- `arithmetic/system13_78.json`, `arithmetic/amalgam14_system67.json`, and associated product/SOS packets: the lower-variable alternatives.
- `low_degree/ordinary15_system61.json` and `low_degree/retained16_system60.json`: full comparison circuits and their single-polynomial forms.
- `low_degree/quartic32.json` and `low_degree/integer_quartic62.json`: all expanded quadratic residuals defining the positive and unrestricted quartics.
- `compiler/tshape_constants.json` and its generator: the complete four-site coefficient recipe and evaluated scalar geometry. The cross version provides a simpler intermediate compiler comparison.
- `proofs/`: derivation notes and independent reviews. `upstream/`: pinned dependencies, machine endpoint, source comparison, and provenance. `SOURCE_PINS.json` and the hash manifest identify the distributed source files.

From the extracted archive directory run

```
python3 verify_arithmetic.py
python3 verify_compiler.py
latexmk -pdf -interaction=nonstopmode -halt-on-error con_zfc_reductions.tex
```

The mathematical checks require only Python's standard library and local files. Typesetting requires a conventional TeX installation with the packages named in the preamble. No network access, external algebra service, giant coefficient materialization, or search for a ZFC refutation is needed.

Operation counts refer to the archived circuits with their four coefficient ports. These ports are bound by the new recipe in (20); they are not universally quantified parameters. Some inherited circuit source packets retain their original coefficient labels for provenance. Renaming their four ports does not alter the formulas or evaluation cost. The manifest distinguishes inherited inputs from newly emitted constructions.

12 Conclusions and sensible next targets

For incorporation, the 13-variable, 84-operation product is the recommended compact equation, with the degree-515 alternative available at one additional operation. If low degree is primary, the 32-positive-variable quartic and 62-integer-variable quartic are the useful outputs. If evaluation cost dominates, the old 67-operation product remains competitive; the new 61-operation system improves its degree without lowering its cost.

The gains arise at different layers and compose cleanly: ordinary strong scaling changes an existential Pell multiplier; weighted root amalgamation removes witnesses while guarding the zero-denominator branch; register elimination lowers the quartic's variable count; translating one local test removes a neighborhood coordinate and shrinks the native compiler. None requires changing the underlying proof-search machine.

Natural further work is to formalize the weighted-root lemma and signed-domain extension, then the three-overlap compiler, before attempting new optimization claims. A search over still

smaller tile presentations or a different exact coefficient layout could improve coefficient height again. This report establishes concrete improvements over the reviewed packages, not a theorem that any one displayed measure is globally minimal.

References

[1] V. Reshetnikov and contributors, *Con(ZFC) in Diophantine form: comparison of the eight report packages*, ProveIt repository, snapshot `3b201cdea32d2c816f5f3d20734051f25fd3ef14`. Pinned comparison and packages 01–08.

[2] ProveIt, package 06, *The 67-operation blank-input polynomial*, including `report.tex`, `proofs/fixed_blank_67_proof.md`, `proofs/pell_lemma_audit.md`, and `code/blank_helical.py`. Pinned package 06.

[3] ProveIt, package 02, `FIXED_INPUT75_ORDINARY_PROOF.md` and `PELL_INTERFACE_LEMMAS.md`. Pinned ordinary-strong package.

[4] ProveIt, `Papers/1980/PELL_RELAXED_AUXILIARY_PROOF.md`, ordinary strong rank lemma and Pell divisibility. Pinned proof note.

[5] ProveIt, `complete84_signed_quotient_absorption.md`, signed auxiliary-root step-down. Pinned signed-quotient note.

[6] ProveIt, `complete84_signed_quotient_soundness_scout.md`, sign-independent forward power, extraction, and population arguments. Pinned soundness note.

[7] ProveIt, `pell_kernel_half_binomial42.md`, strong half-binomial Pell kernel. Pinned kernel note.

[8] ProveIt, package 06, `proofs/blank_helical_proof.md`, finite-strip soundness and padded helical completeness. Pinned compiler proofs.

[9] ProveIt, `KEndpoint215.lean` and the K215 verification package, `blank_halts_iff_zfc_refutation` and `nonhalting_iff_zfc_consistency`. Pinned K215 verification. The local endpoint source is included in `upstream/logic`.

[10] ProveIt, `Diophantine/Paper1982/NineMain.lean`, `Jones1982.nine_unknowns`, and its formalized dependencies. Pinned nine-unknowns existence theorem.